Chapter 3 Multidimensional Random Variables

Section 1 Two-Dimensional Random Variables

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Outline

1. Two-Dimensional Random Variables
   - Definition of Two-Dimensional Random Variables
   - Definition of the Distribution Function
   - Examples
   - Two-Dimensional Discrete Random Variables
   - Examples (Discrete Case)
   - Two-Dimensional Continuous Random Variables
   - Uniform Distribution
   - Normal Distribution

Why do we need two-dimensional random variables?

- Measuring a person's height X and weight Y;
- Recording the highest temperature X and lowest temperature Y in a region;
- Tossing two coins, recording the number of heads X and Y;
- Investigating the relationship between a father's height X and his son's height Y.

Common Feature

Multiple random variables are defined on the same sample space, and they are often interconnected.
To study the probability laws of "multiple random variables taking values together," we need to introduce **two-dimensional random variables**.

Definition: Two-Dimensional Random Variable

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Let X and Y be two random variables defined on the same sample space Ω. The ordered pair

$$(X, Y)$$

is called a **two-dimensional random variable** (or **two-dimensional random vector**).
Generally, let $X_1, X_2, \ldots, X_n$ be n random variables defined on the same sample space Ω. The ordered tuple

$$(X_1, X_2, \ldots, X_n)$$

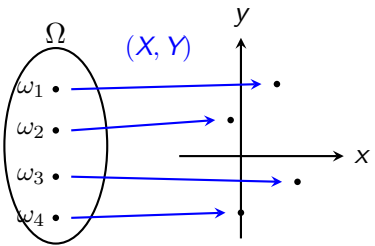
is called an n-**dimensional random variable** (or n-**dimensional random vector**).

Intuition: From Sample Space to the Plane

Mapping Perspective

$$(X, Y) : \Omega \to \mathbb{R}^2, \quad \omega \mapsto (X(\omega), Y(\omega)).$$

A two-dimensional random variable is essentially a "vector-valued function." Studying its distribution means studying the probability of it falling into various regions in the plane.



Definition: Joint Distribution Function

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Let (X, Y) be a two-dimensional random variable. For any real numbers x, y, the function

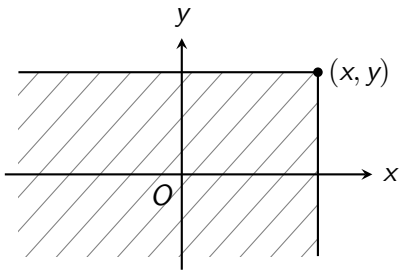
$$F(x, y) = P(X \leq x, \ Y \leq y)$$

is called the **distribution function** of (X, Y), or the **joint distribution function** of X and Y.

Geometric Meaning

$F(x, y)$ represents the probability that the random point (X, Y) falls into the planar region $(-\infty, x] \times (-\infty, y]$.

Illustration of the Joint Distribution Function



$F(x, y) = P(X \leq x,\ Y \leq y)$: Probability of the shaded region extending infinitely to the bottom-left.

Understanding via Inverse Mapping

Key Requirement

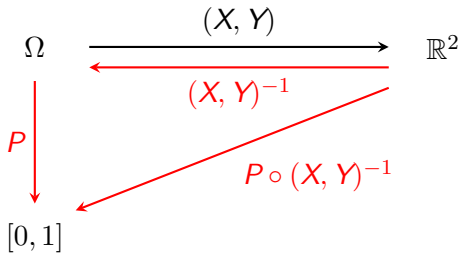
For any set $B \subset \mathbb{R}^2$ in the plane, its pre-image

$$(X, Y)^{-1}(B) = \{\omega \in \Omega : (X(\omega), Y(\omega)) \in B\}$$

must be an event (i.e., belongs to the σ-algebra $\mathcal{F}$).

The role of the two-dimensional random variable (X, Y) is to "pull back" a set B from the plane to the sample space, yielding the event $(X, Y)^{-1}(B)$. This allows us to study random phenomena using set operations and analytical tools in the plane.

Inverse Mapping Diagram



$$P \circ (X, Y)^{-1}(B) = P\{\omega : (X(\omega), Y(\omega)) \in B\}$$

Inverse Mapping Interpretation of the Distribution Function

Take the set in the plane:

$$B = (-\infty, x] \times (-\infty, y],$$

then

$$(X, Y)^{-1}(B) = \{\omega : X(\omega) \leq x,\ \ Y(\omega) \leq y\} = \{X \leq x,\ \ Y \leq y\}.$$

Therefore,

$$F(x, y) = P(X \leq x,\ \ Y \leq y) = P\big((X, Y)^{-1}((-\infty, x] \times (-\infty, y])\big).$$

Conclusion

The joint distribution function is essentially "the probability of the inverse image of the planar set $(-\infty, x] \times (-\infty, y]$."

Proposition: Properties of the Joint Distribution Function

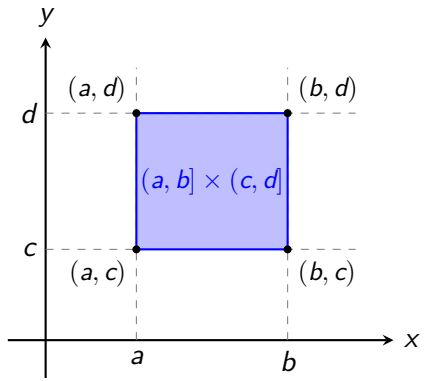
Let $F(x, y)$ be the joint distribution function of (X, Y). Then:
(1) **Monotonicity**: $F(x, y)$ is non-decreasing in both x and y.
(2) **Boundedness**: $0 \leq F(x, y) \leq 1$, and

$$\lim_{x \to -\infty} F(x, y) = 0, \quad \lim_{y \to -\infty} F(x, y) = 0, \quad \lim_{x \to +\infty,\ y \to +\infty} F(x, y) = 1.$$

(3) **Right-continuity**: $F(x, y)$ is right-continuous in x and in y.
(4) **Rectangle Probability Formula**: For any $a < b$, $c < d$,

$$P(a < X \leq b,\ c < Y \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c).$$

Illustration of the Rectangle Probability Formula



$$P = F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

Proof Idea for the Rectangle Probability Formula

The event $\{a < X \leq b,\ c < Y \leq d\}$ corresponds to the rectangle $(a, b] \times (c, d]$. Using the definition of the distribution function,

$$\begin{aligned} P(a < X \leq b,\ c < Y \leq d) &= P(X \leq b,\ Y \leq d) - P(X \leq a,\ Y \leq d) \\ &\quad - P(X \leq b,\ Y \leq c) + P(X \leq a,\ Y \leq c) \\ &= F(b, d) - F(a, d) - F(b, c) + F(a, c). \end{aligned}$$

Remark

This is the two-dimensional generalization of the 1D formula $P(a < X \leq b) = F(b) - F(a)$.

Example: Distribution Function of a Discrete RV

Example: Joint CDF of a Discrete Random Variable

Toss two fair coins. Let

$$X = \{\text{Number of heads on the first coin}\}, \quad Y = \{\text{Number of heads on the second coin}\}.$$

The joint PMF is given by

$$P(X = 0, Y = 0) = P(X = 0, Y = 1) = P(X = 1, Y = 0) = P(X = 1, Y = 1) = \frac{1}{4}.$$

Find the joint distribution function $F(x, y)$ of (X, Y).

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Analysis Hints

- List all possible values: $(0, 0), (0, 1), (1, 0), (1, 1)$, each with probability $\frac{1}{4}$;
- Use the lines $x = 0, x = 1, y = 0, y = 1$ to partition the plane;
- In each region, sum the probabilities of values satisfying $x_i \leq x$ and $y_j \leq y$;
- Note right-continuity: $x = 0$ or $y = 0$ should be included on the "inclusive" side.

Example (Continued): Solution and Result

$$F(x,y) = \begin{cases} 0, & x < 0 \text{ or } y < 0, \\ \frac{1}{4}, & 0 \le x < 1,\ 0 \le y < 1, \\ \frac{1}{2}, & 0 \le x < 1,\ y \ge 1, \\ \frac{1}{2}, & x \ge 1,\ 0 \le y < 1, \\ 1, & x \ge 1,\ y \ge 1. \end{cases}$$

Observations

- It is a **right-continuous two-dimensional step function**;
- The jump points are exactly $(0,0), (0,1), (1,0), (1,1)$;
- The jump height at each point equals the corresponding probability $\frac{1}{4}$.

Example: Uniform Distribution on a Square

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Let (X, Y) be uniformly distributed on the square region $D = [0, 2] \times [0, 2]$, i.e.,

$$P\big((X, Y) \in A\big) = \frac{m(A)}{m(D)}, \qquad A \subset D.$$

Find the joint distribution function $F(x, y)$.

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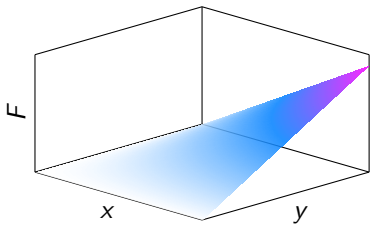
Analysis Hints

- $F(x, y)$ equals the area of the intersection of $(-\infty, x] \times (-\infty, y]$ with D, divided by $m(D) = 4$;
- Partition into five cases based on the position of (x, y) relative to the square;
- Inside the square: the intersection is $[0, x] \times [0, y]$ with area xy.

Example (Continued): Solution

$$F(x,y) = \begin{cases} 0, & x < 0 \text{ or } y < 0, \\ \dfrac{xy}{4}, & 0 \le x \le 2,\ 0 \le y \le 2, \\ \dfrac{y}{2}, & x > 2,\ 0 \le y \le 2, \\ \dfrac{x}{2}, & 0 \le x \le 2,\ y > 2, \\ 1, & x > 2,\ y > 2. \end{cases}$$

Inside D, $F(x,y) = \dfrac{xy}{4}$ is a bilinear function.



Definition: Discrete RV and Joint PMF

Definition: Two-Dimensional Discrete Random Variable

Let (X, Y) be a two-dimensional random variable. If both X and Y are discrete random variables, then (X, Y) is called a **two-dimensional discrete random variable**.

Definition: Joint Probability Mass Function (PMF)

Let (X, Y) be a two-dimensional discrete random variable with all possible values (x_i, y_j), $i = 1, 2, \ldots$, $j = 1, 2, \ldots$. The function

$$p_{ij} = P(X = x_i, \ Y = y_j)$$

is called the **joint probability mass function** (or **joint PMF**) of (X, Y).

Properties and Tabular Representation of the Joint PMF

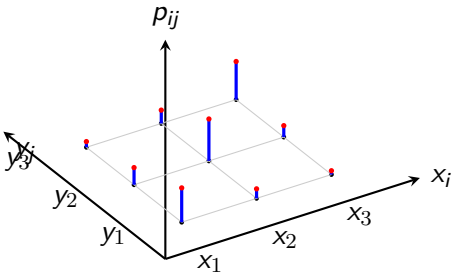
Two Basic Properties

(1) **Non-negativity**: $p_{ij} \geq 0$;

(2) **Normalization**: $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} p_{ij} = 1.$

$X \backslash Y$	y_1	y_2	$\cdots$	y_j	$\cdots$
x_1	p_{11}	p_{12}	$\cdots$	p_{1j}	$\cdots$
x_2	p_{21}	p_{22}	$\cdots$	p_{2j}	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	
x_i	p_{i1}	p_{i2}	$\cdots$	p_{ij}	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	

Vertical Line Chart of the Joint PMF



Vertical line length $= p_{ij}$, sum of all lengths $= 1$.

Example: Joint PMF for Tossing Two Coins

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Toss two fair coins. Let

$$X = \{\text{Number of heads on the first coin}\}, \quad Y = \{\text{Number of heads on the second coin}\}.$$

Find the joint PMF of (X, Y).

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Find the joint PMF of (X, Y).

Analysis Hints

- X, Y only take values $0, 1$. The two coins do not affect each other $\Rightarrow$ independent;
- $P(X = i, Y = j) = P(X = i)P(Y = j)$;
- Marginals: $P(X = 0) = P(X = 1) = P(Y = 0) = P(Y = 1) = \frac{1}{2}$.

Example (Continued): Result

$$P(X=i, Y=j) = P(X=i)P(Y=j) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}, \quad i,j = 0,1.$$

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$

Example: Drawing Balls Without Replacement

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A bag contains 2 red balls and 3 white balls. Two balls are drawn successively without replacement. Let

$$X = \begin{cases} 1, & \text{1st ball is red,} \\ 0, & \text{1st ball is white,} \end{cases} \qquad Y = \begin{cases} 1, & \text{2nd ball is red,} \\ 0, & \text{2nd ball is white.} \end{cases}$$

Find the joint PMF of (X, Y).

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Find the joint PMF of (X, Y).

Analysis Hints

- Four possible values: $(0,0), (0,1), (1,0), (1,1)$;
- Multiplication rule: $P(X = i, Y = j) = P(X = i)\, P(Y = j \mid X = i)$;
- For the second draw, only 4 balls remain, and the number of red balls depends on the first draw.

Example (Continued): Solution

$$P(X=1,Y=1)=\frac{2}{5}\cdot\frac{1}{4}=\frac{1}{10}, \qquad P(X=1,Y=0)=\frac{2}{5}\cdot\frac{3}{4}=\frac{3}{10},$$

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$X \backslash Y$	0	1
0	$\frac{3}{10}$	$\frac{3}{10}$
1	$\frac{3}{10}$	$\frac{1}{10}$

Definition: Continuous RV and Joint Density

Definition: Two-Dimensional Continuous Random Variable

Let (X, Y) be a two-dimensional random variable with joint distribution function $F(x, y)$. If there exists a non-negative integrable function $f(x, y)$ such that for any real numbers x, y,

$$F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, dv\, du,$$

then (X, Y) is called a **two-dimensional continuous random variable**, and $f(x, y)$ is called the **joint probability density function** (or **joint density**) of (X, Y).

Properties of the Joint Density Function

Four Basic Properties

(1) **Non-negativity**: $f(x, y) \geq 0$;

(2) **Normalization**:

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y)\, dx\, dy = 1;$$

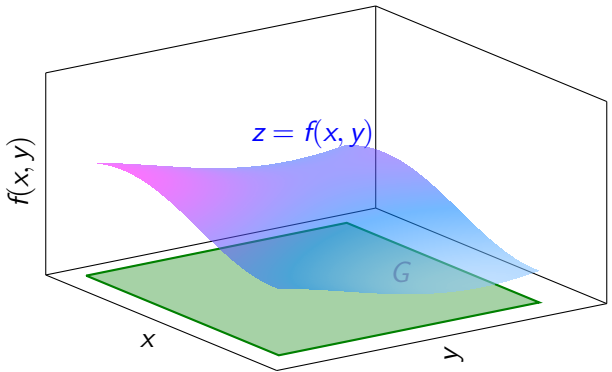
(3) **Region Probability Formula**: For any region G in the plane,

$$P\big((X, Y) \in G\big) = \iint_G f(x, y)\, dx\, dy;$$

(4) **Relationship with CDF**: At points where f is continuous,

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y).$$

Geometric Meaning of Region Probability



$P((X, Y) \in G) = \iint_G f(x, y)\, dx\, dy$: Volume of the solid under the surface.

Example: Uniform Distribution on a Rectangle

Example: Two-Dimensional Uniform Distribution

Let (X, Y) be uniformly distributed on the rectangle $D = [a, b] \times [c, d]$. Find the joint density function.

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Let (X, Y) be uniformly distributed on the rectangle $D = [a, b] \times [c, d]$. Find the joint density function.

Analysis Hints

- Uniform distribution $\Rightarrow$ density is constant C on D, and 0 outside D;
- Normalization: $\iint_D C \, dx \, dy = C \cdot m(D) = 1$;
- Solve for $C = 1/m(D)$.

Example (Continued): Result and General Case

$$f(x,y) = \begin{cases} \dfrac{1}{(b-a)(d-c)}, & a \le x \le b,\ c \le y \le d, \\ 0, & \text{otherwise.} \end{cases}$$

More Generally

If (X, Y) is uniformly distributed on a planar region D, then

$$f(x,y) = \begin{cases} \dfrac{1}{m(D)}, & (x,y) \in D, \\ 0, & \text{otherwise,} \end{cases}$$

where $m(D)$ denotes the area of region D.

Definition: Bivariate Normal Distribution

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If the joint density function of (X, Y) is

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right\},$$

where $\mu_1, \mu_2 \in \mathbb{R}$, $\sigma_1, \sigma_2 > 0$, $-1 < \rho < 1$, then (X, Y) follows a **bivariate normal distribution**, denoted as

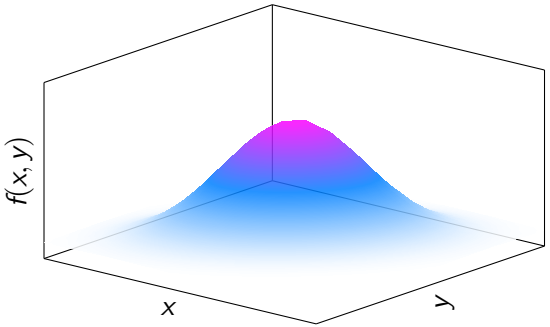
$$(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho).$$

Bivariate Normal Distribution: Parameters and Shape

Parameter Meaning

- (μ_1, μ_2): Center of the density peak;
- σ_1, σ_2: Spread in the two respective directions;
- ρ: **Correlation coefficient**, characterizing the degree of linear correlation between X and Y.

When $\rho = 0$, X and Y are uncorrelated; in the bivariate normal case, $\rho = 0$ is also equivalent to X and Y being independent.



Observations on the Bivariate Normal Density Surface

- The shape of the density function resembles a "hat";
- The contour lines are a family of ellipses;
- The center of the ellipses is (μ_1, μ_2);
- The shape of the ellipses is jointly determined by σ_1, σ_2, and ρ.

Section Summary

- **Two-Dimensional Random Variables**: An ordered pair of random variables (X, Y) on the same sample space, viewed as a mapping $\Omega \to \mathbb{R}^2$;
- **Joint Distribution Function**: $F(x, y) = P(X \leq x, Y \leq y)$, possessing monotonicity, boundedness, right-continuity, and the rectangle probability formula;
- **Two-Dimensional Discrete**: Joint PMF p_{ij}, non-negative and normalizing to 1, representable via tables or vertical line charts;
- **Two-Dimensional Continuous**: Joint density $f(x, y)$, where region probabilities are double integrals, and $\partial^2 F/\partial x \partial y = f$;
- **Common Distributions**: Uniform distribution on a rectangle/region; Bivariate normal distribution $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$.